

General Clerkship

Whole Class Session

Introduction to CVS Investigations

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Why do you want to perform *history taking* and *physical examination*?

Mr Poon

- M/38
- Good past health
- 1 month history of progressive exercise intolerance
- What do you want to know?

- **Reason for exercise intolerance**

?Heart Failure; Aortic Regurgitation

- E.g. orthopnea, paroxysmal nocturnal dyspnea, lower limb swelling; pitting edema, ↑JVP, collapsing pulse, Corrigan sign, deviated apex, EDM 2/6 at LSB, bilateral lower zone crepitations

- **Underlying cause (predisposing factors and precipitating factors)**

?Marfan Syndrome; Aortic Disease

- E.g. Father died of aortic rupture at late 40s; tall, long and slender fingers, pectus excavatum, wrist sign and thumb sign

- **Severity / complications**

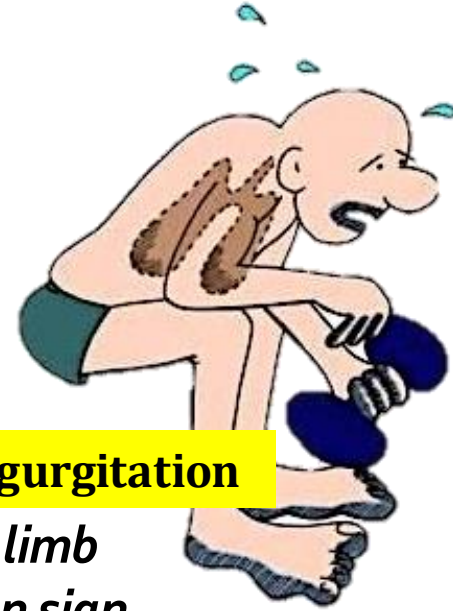
?Respiratory Failure

- E.g. Cyanotic: requiring admission for oxygen / ventilation

- **Management**

?Aortic dissection and Aortic aneurysm

- E.g. no pulse deficit, no pulsatile mass on abdominal exam



Messages

- The aims of the investigations include confirming the diagnosis, delineating the underlying cause, assessing severity and complications, and obtaining essential information required for the successful management of the patient
- Investigations should be based on initial *differential diagnoses*
- *Do not wait for investigations during emergency!*

Common Cardiac Symptoms

- **Dyspnea**
- **Chest pain**
- **Lower limb swelling**
- **Palpitation**
- **Syncope / presyncope**

Etiologies

Dyspnea

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graph TD; A[Dyspnea] --> B[Cardiovascular disorders]; A --> C[Respiratory disorders]; A --> D[Others];
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Cardiovascular disorders

(e.g. heart failure, coronary artery disease, pulmonary embolism, pulmonary hypertension)

Respiratory disorders

(e.g. pneumonia, pneumonitis, COPD, asthma, bronchiectasis, interstitial lung disease)

Others

(e.g. anemia, metabolic acidosis, myopathy, thyrotoxicosis, hyperventilation syndrome)

Definition of Heart Failure

HF is a clinical syndrome consisting of cardinal symptoms (*e.g. dyspnea, ankle swelling, fatigue*) and signs (*e.g. elevated JVP, pulmonary crackles, peripheral edema*) due to any structural (*e.g. cardiomyopathies*) or functional (*e.g. tachy- or brady-arrhythmias*) abnormalities of the cardiovascular system that results in elevated intracardiac pressures and / or inadequate cardiac output at rest and / or during exercise

Common Investigations for Initial Evaluation

- Complete blood count
- Serum electrolyte (Na, K, Ca, PO₄), urea, creatinine, glucose
- pH, PO₂, PCO₂, HCO₃⁻
- Markers for infection / inflammation (e.g. CRP C-reactive protein, PCT procalcitonin)
- Chest x-ray
- ECG
- High-sensitive troponin
- BNP brain natriuretic peptide / NT pro BNP
- D-dimer
- Thyroid-stimulating hormone, free T₄
- Urine for protein, blood, and glucose, microscopic urinalysis

- *Should refine your list of investigations based on your list of differential diagnoses derived from history and physical examination!!!*
- *E.g. If this is a patient with low grade fever and new murmur, you will order repeated blood smear and culture to exclude infective endocarditis immediately!*

BNP / NT pro BNP

- Brain natriuretic peptide (BNP) / N-terminal pro brain natriuretic peptide (NT pro BNP)
- BNP are released from cardiomyocytes (primarily those in ventricles) in response to wall stretch
- Diuretic, natriuretic and antihypertensive effect by inhibiting the renin-angiotensin-aldosterone system
- proBNP is the precursor of BNP and NT pro BNP
- Plasma half-life
 - BNP: 20 min; NT pro BNP: 60-120 min
- Storage at room temperature
 - BNP: 4 hours; NT-proBNP: 72 hours
- Higher than age-specific reference range *rule in* HF
- *Low BNP or NT-proBNP have a very high negative predictive value to rule out HF*

Alternative Causes of ↑BNP / NT pro BNP

Cardiac

Heart failure

ACS

Pulmonary embolism

Myocarditis

Left ventricular hypertrophy

Hypertrophic or restrictive cardiomyopathy

Valvular heart disease

Congenital heart disease

Atrial and ventricular tachyarrhythmias

Heart contusion

Cardioversion, ICD shock

Surgical procedures involving the heart

Pulmonary hypertension

Non-Cardiac

Advanced age

Ischaemic stroke

Subarachnoid haemorrhage

Renal dysfunction

Liver dysfunction (mainly liver cirrhosis with ascites)

Paraneoplastic syndrome

COPD

Severe infections (including pneumonia and sepsis)

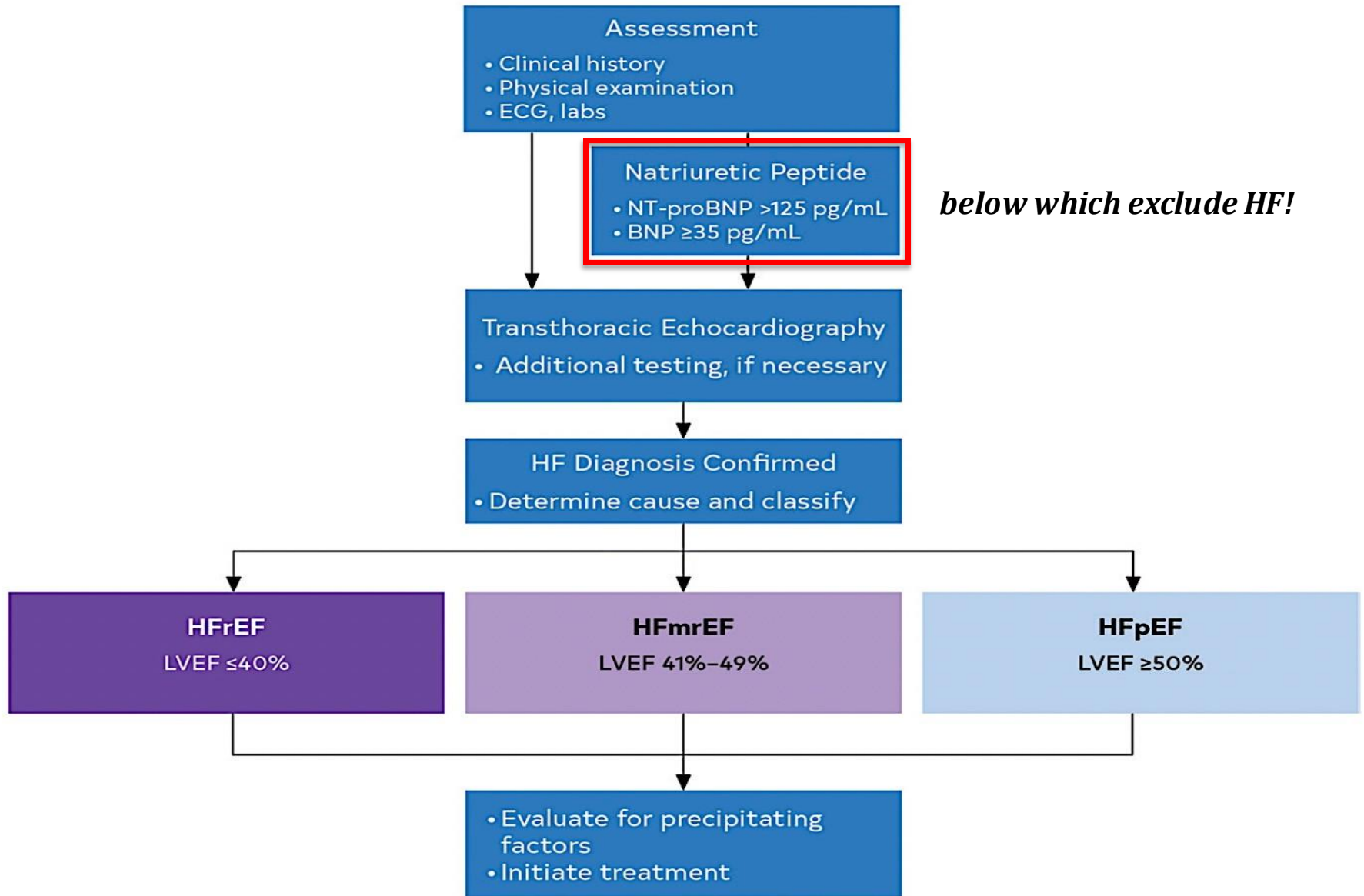
Severe burns

Anaemia

Severe metabolic and hormone abnormalities
(e.g. thyrotoxicosis, diabetic ketosis)

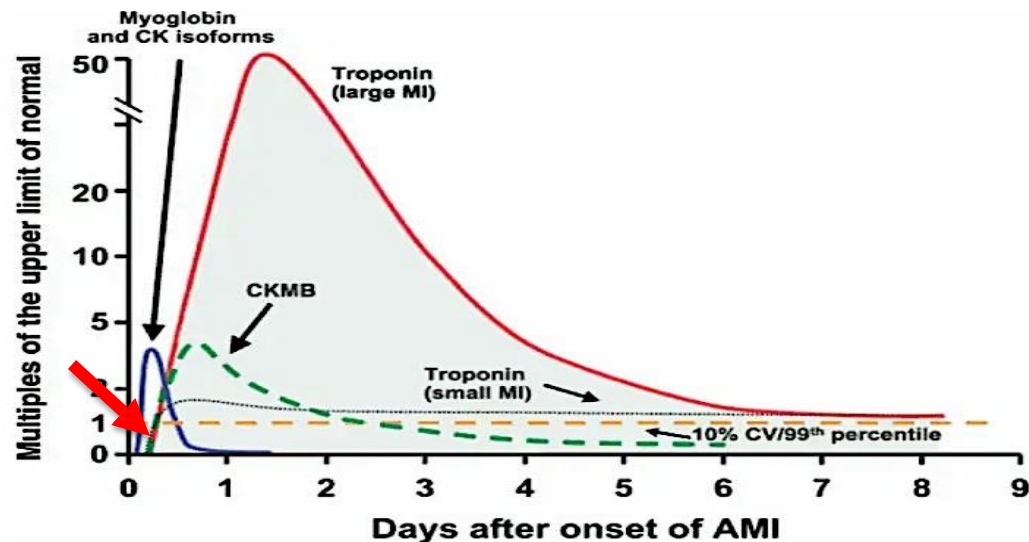
***Obesity – falsely low BNP / NT pro BNP due to
↓NPRA/NPRC, ↑neprilysin and ↑renal filtration***

Diagnostic Algorithm for Patients With Suspected HF

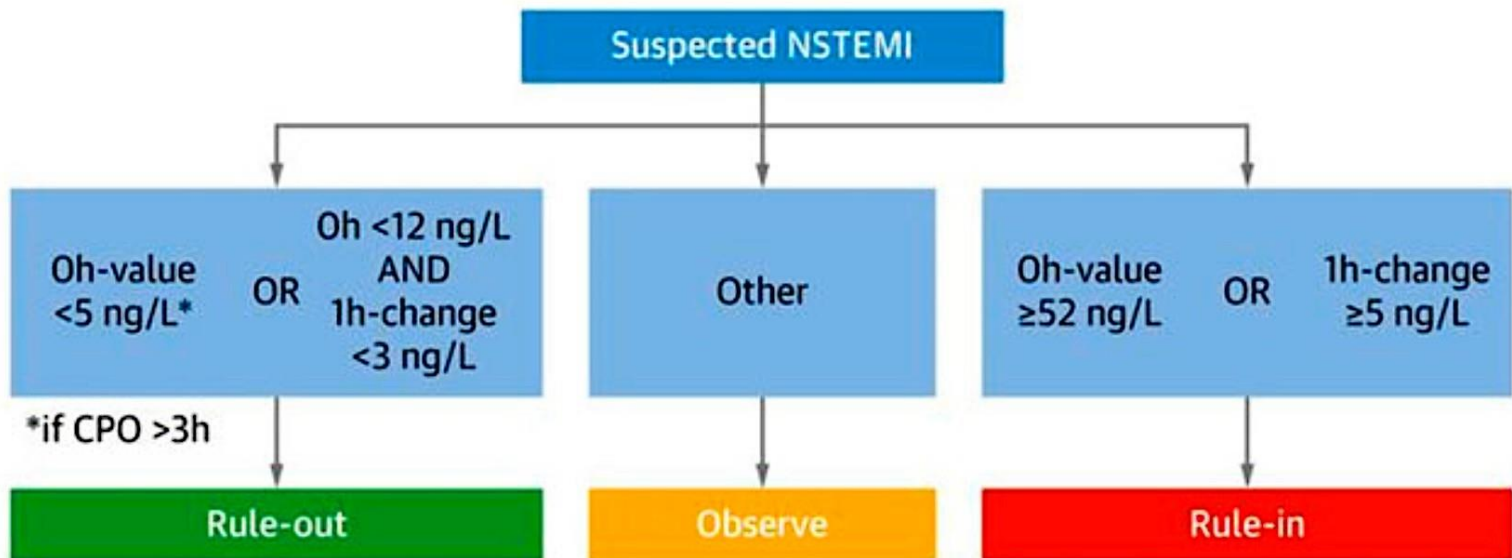
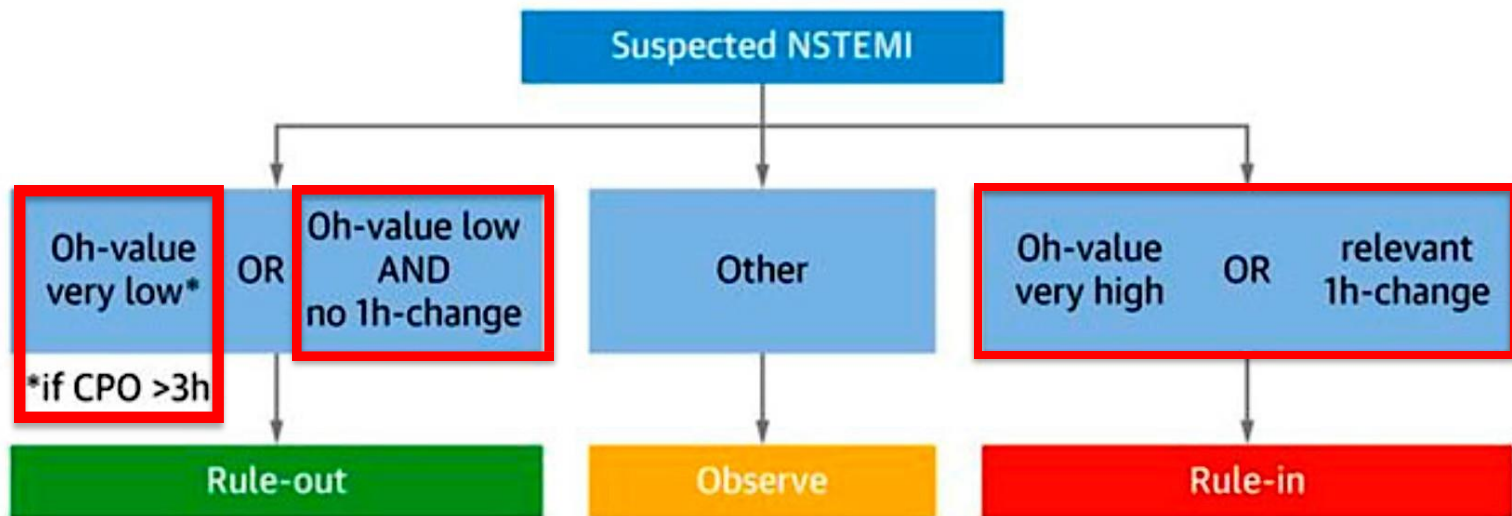


Troponin

- Troponin are protein complexes that modulate contraction and relaxation of striated muscles
- 7% of cardiac troponin T (cTnT) and 3.5% of cTnI are found freely in cardiac myocyte cytoplasm; fragments cleared by kidneys
- cTnT and cTnI are specific markers for myocardial injury
- *High sensitive* troponin (hs-cTn) refers to *analytical sensitivity*
 - Same cardiac troponin
 - Much lower limits of detection but no effect on 99th percentile cut-off point for myocardial injury
 - Enable earlier decision-making by measuring both absolute value and comparing values at 1 h / 2h with initial value (1- or 2-h hs-cTn Δ)



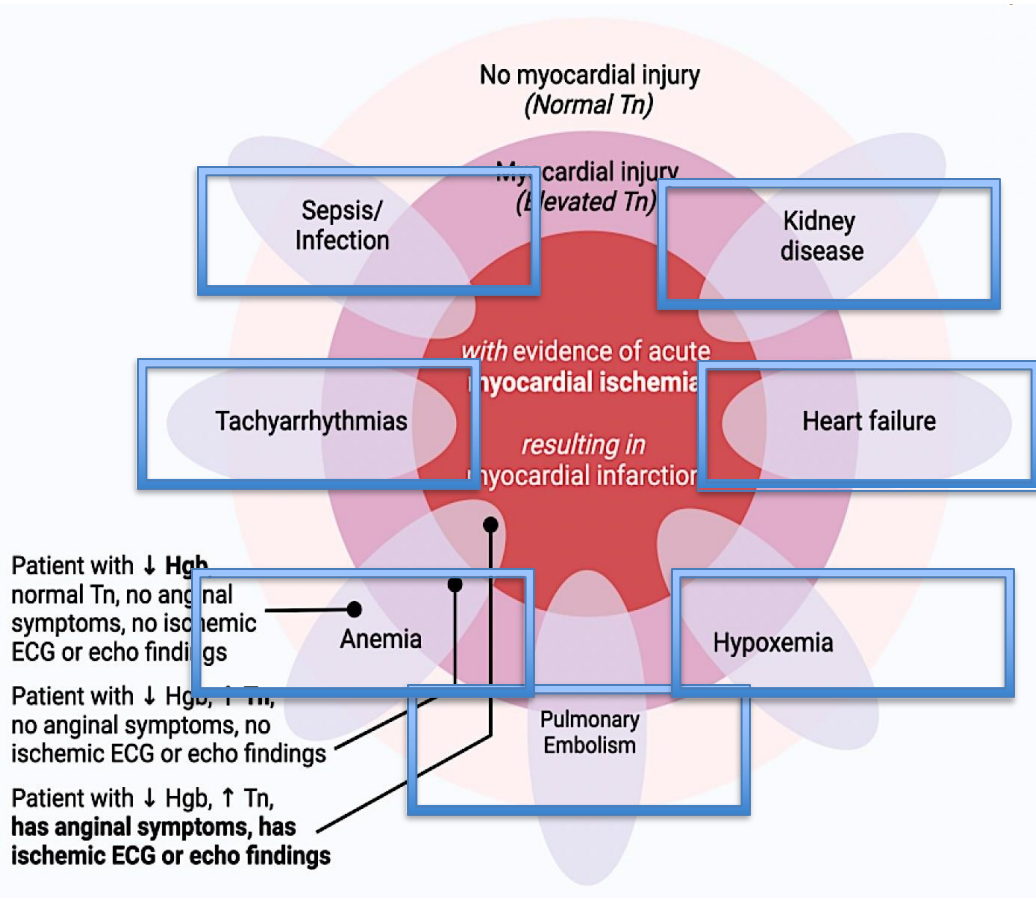
ESC 0/1h Rule for Suspected ACS



Alternative Causes of ↑Troponin

True Myocardial Injury

True False Positive



1. *Heterophilic antibodies*
2. *Severe skeletal muscle disease e.g. Duchenne muscular dystrophy / rhabdomyolysis*

D-Dimer

- D-dimer is the degradation product of crosslinked fibrin
- Reflects ongoing activation of the hemostatic system
- Cleared by kidneys and reticulo-endothelial system
- Plasma half-life ~ 8 h
- ELISA / ELISA-derived assays have a diagnostic sensitivity of $\geq 95\%$ for hemodynamically stable pulmonary embolism
- Standard cut-off: *500 ug/L*
- Age-adjusted cut-off: *age x 10 ug/L for those >50 yrs*
- Increased in the following populations:
 - Cancer
 - Severe infection / inflammation
 - Pregnancy
 - Hospitalization

Suspected PE in a patient without haemodynamic instability^a

Assess clinical probability of PE

Clinical judgement or prediction rule^b

Low or intermediate clinical probability,
or PE unlikely

High clinical probability
or PE likely

D-dimer test

Negative

Positive

CTPA

CTPA

No PE

PE confirmed^d

No PE

PE confirmed^d

No treatment^c

Treatment^c

No treatment^c
or investigate
further^e

Treatment^c

Revised Geneva Clinical Prediction Rule

Previous PE or DVT	1
Heart rate	
75–94 b.p.m.	1
≥95 b.p.m.	2
Surgery or fracture within the past month	1
Haemoptysis	1
Active cancer	1
Unilateral lower-limb pain	1
Pain on lower-limb deep venous palpation and unilateral oedema	1
Age >65 years	1

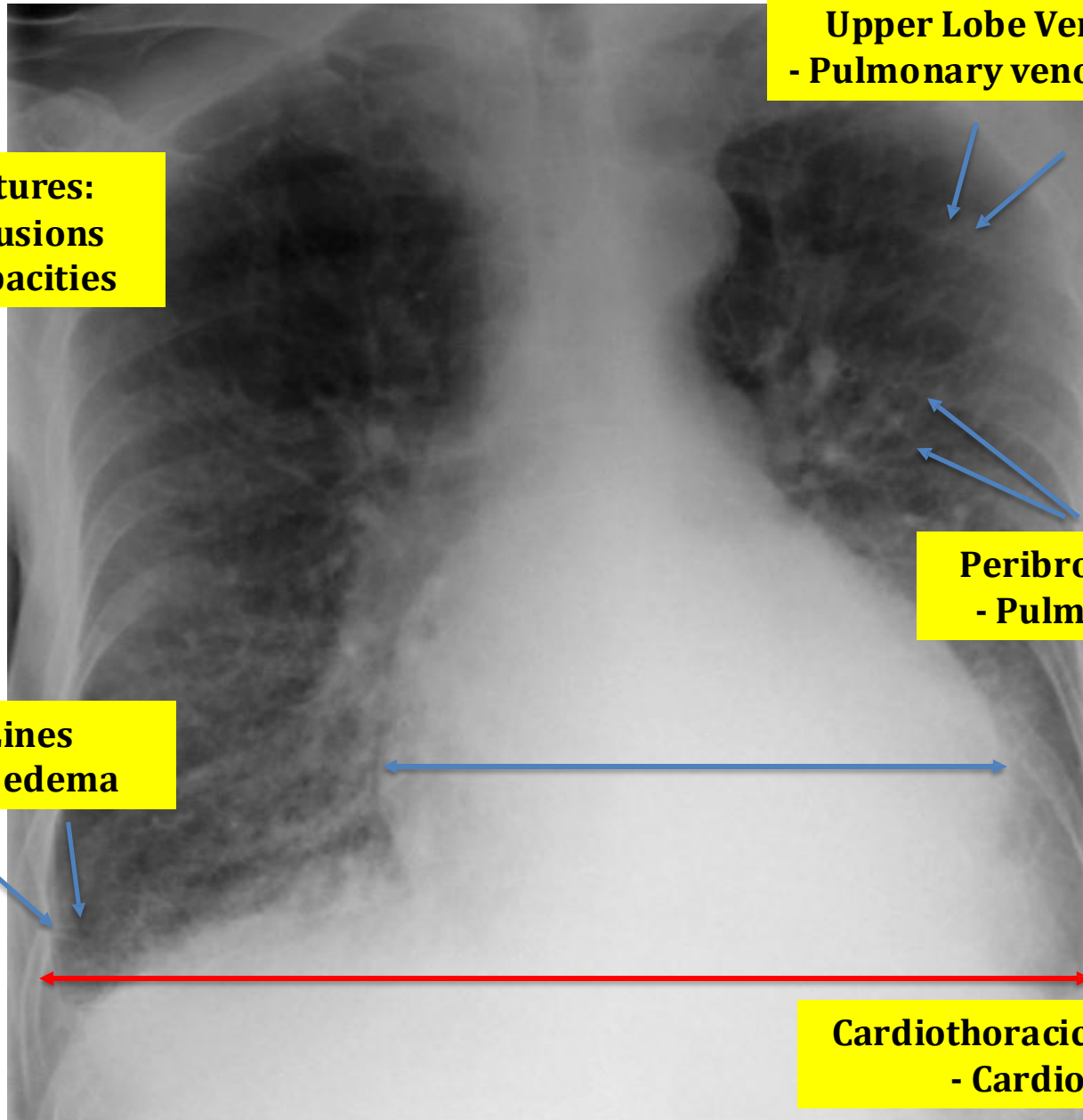
Clinical probability

Three-level score	
Low	0–1
Intermediate	2–4
High	≥5
Two-level score	
PE-unlikely	0–2
PE-likely	≥3

Chest X ray

- **Heart Failure**
- **Aortic Dissection / Aneurysm**
- **Primary Respiratory Diseases**
- **Structural Heart Diseases**
- **Pulmonary Embolism**
- **Pulmonary Hypertension**

Heart Failure



Upper Lobe Venous Diversion
- Pulmonary venous hypertension

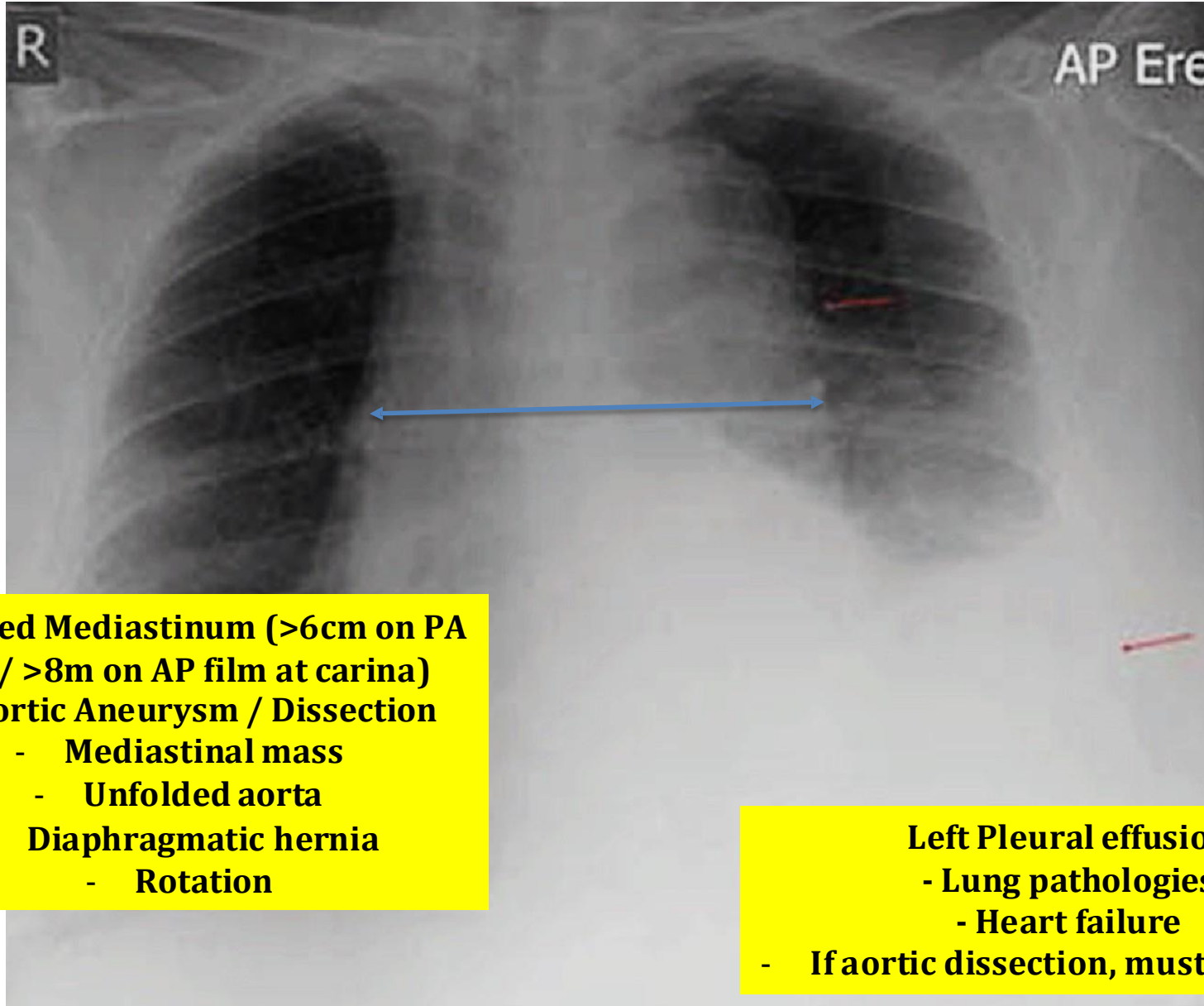
Other Features:
Pleural effusions
Bat wing opacities

Peribronchovascular Cuffing
- Pulmonary edema

Kerley B Lines
- Pulmonary edema

Cardiothoracic Ratio ($N < 0.5$)
- Cardiomegaly

Aortic Dissection



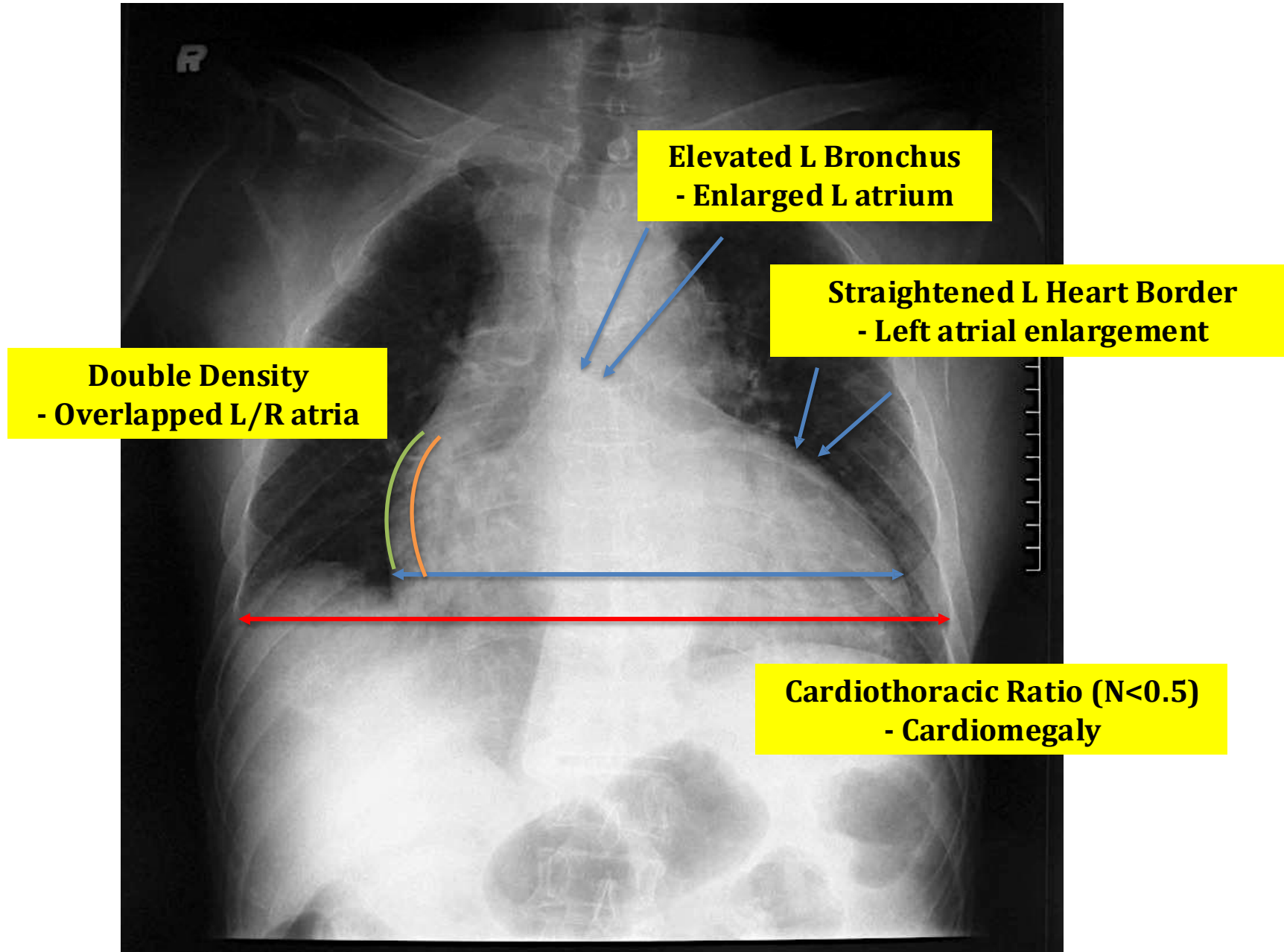
Widened Mediastinum (>6cm on PA film / >8cm on AP film at carina)

- Aortic Aneurysm / Dissection
 - Mediastinal mass
 - Unfolded aorta
- Diaphragmatic hernia
 - Rotation

Left Pleural effusion

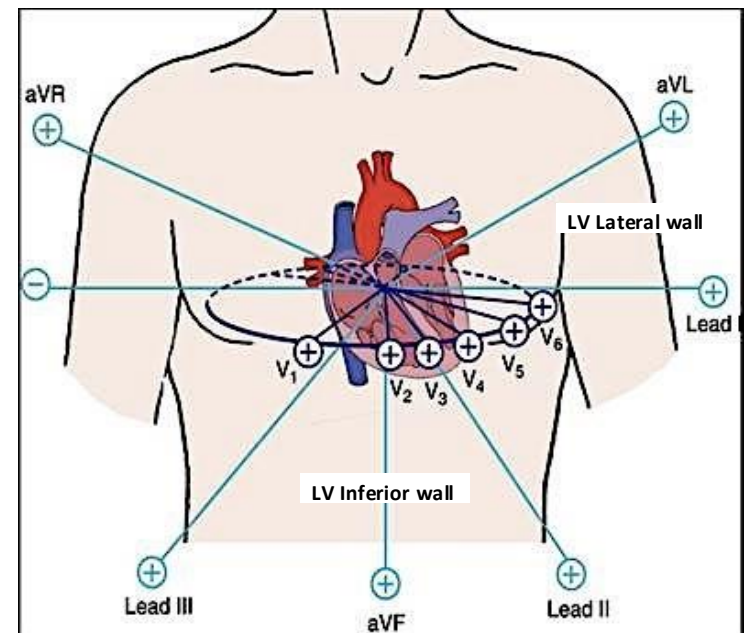
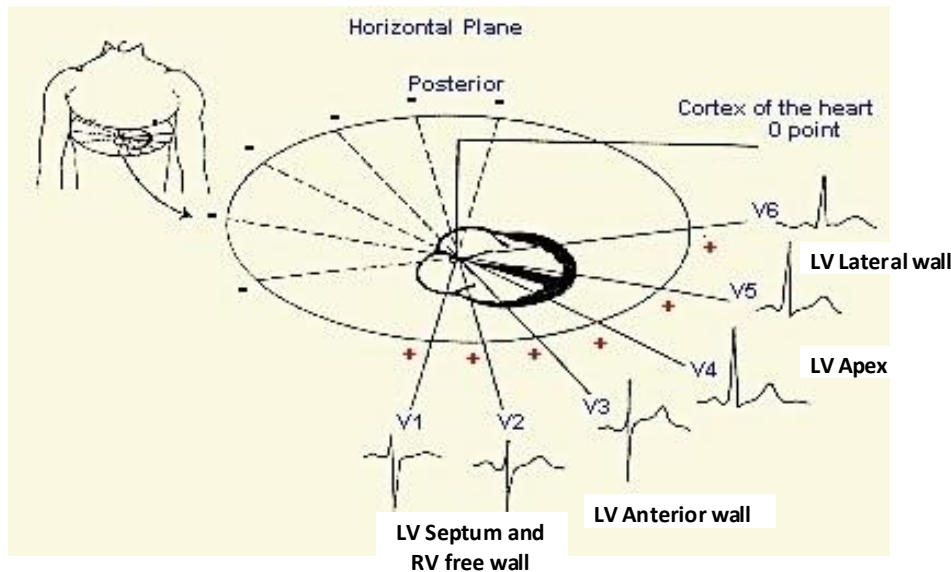
- Lung pathologies
- Heart failure
- If aortic dissection, must be type A!

Mitral Stenosis



ECG

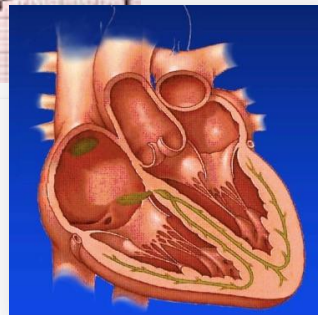
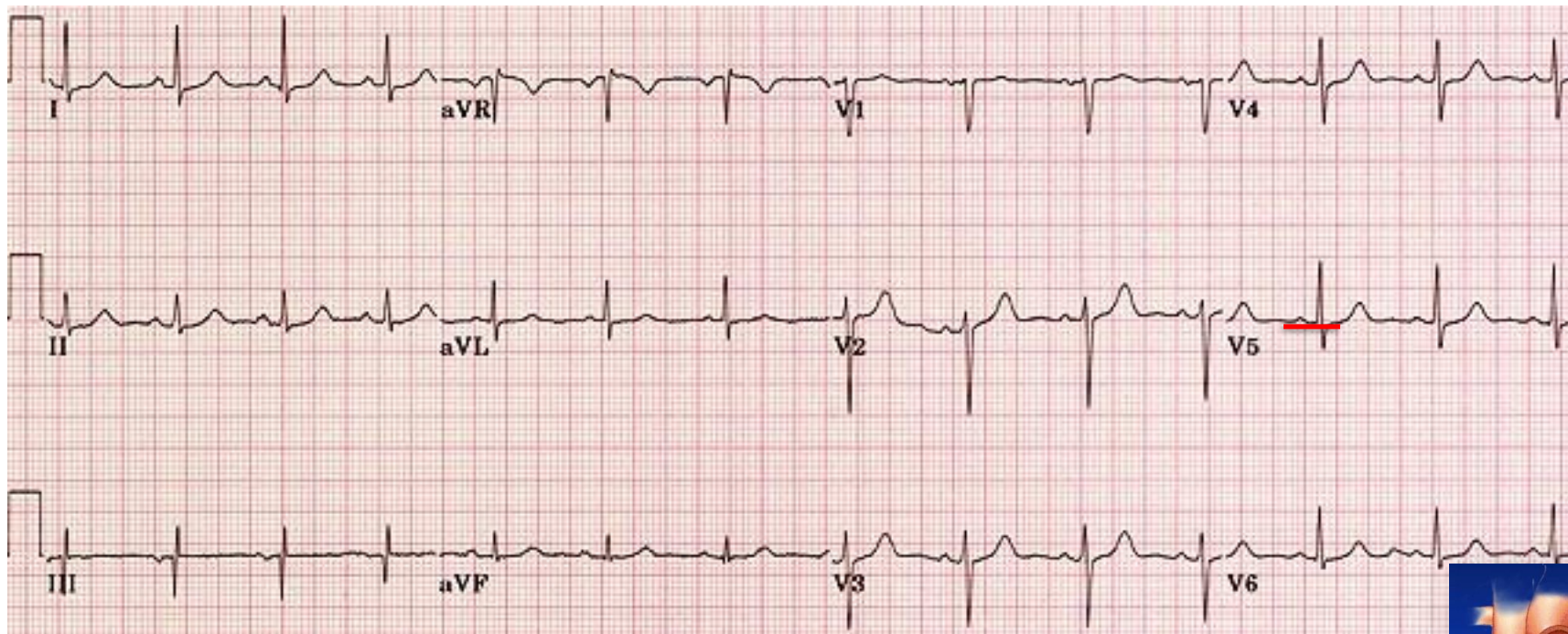
- Recording of the heart's electrical activity from electrodes on the body surface
- Initial investigations of ALL potential cardiac conditions!
- Systemic evaluation for all intervals, rhythm, voltage and waveforms
- Understand the localizing capability of ECG leads



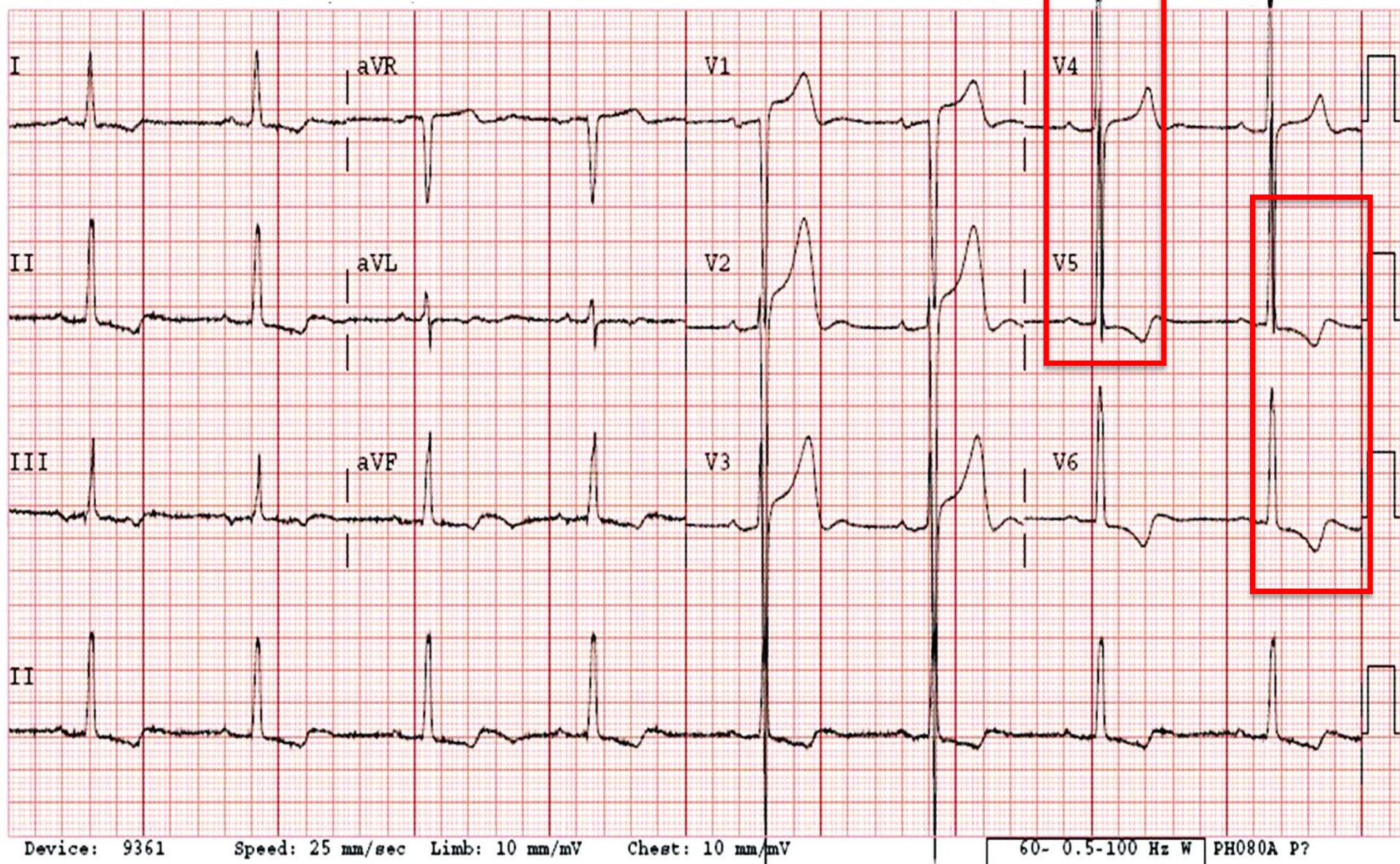
12-lead ECG Changes in Common Cardiac Diseases

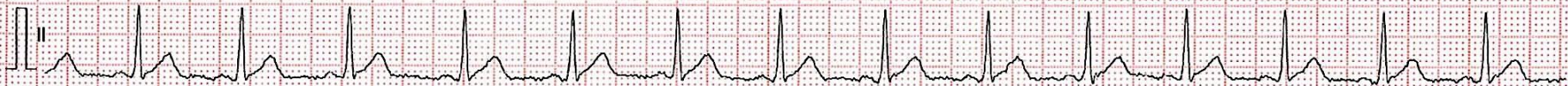
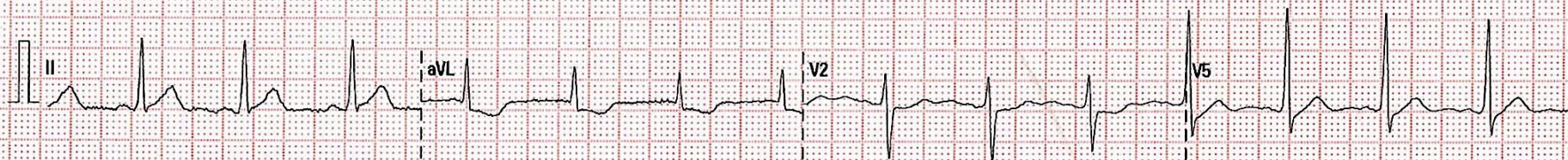
Conditions	ECG changes
Cardiomyopathies	LV hypertrophy, low voltage, poor R progression, bundle branch block, Q waves, ST depression, T wave inversion, epsilon wave, AF, PVCs / NSVT
Coronary artery disease	Regional upwardly convex / straight pattern ST elevation, ST depression, T wave inversion, Q waves, bundle branch block (LBBB!), PVCs / NSVT
Arrhythmia	Sinus bradycardia, 1 st , 2 nd and 3 rd degree AV block; bundle branch block; sinus tachycardia, supraventricular tachycardia, AF / flutter, PACs; PVCs / NSVT, delta wave, QTc interval, Brugada pattern
Pericardial diseases	Widespread upwardly concave ST elevation with PR depression (reciprocal changes in aVR possible), T wave flattened later followed by inversion, low voltage
Pulmonary embolism	Sinus tachycardia, S1Q3T3, complete / incomplete RBBB, right axis deviation, dominant R wave V1, T wave inversion precordial lead, delayed precordial transition
Myocarditis	Sinus tachycardia, non-specific ST/T changes, prolonged QRS / QTc, bundle branch block, progressive AV block, PVC / NSVT, can be associated with pericarditis
Other examples	Amyloidosis: low voltage; Sarcoidosis: progressive AV block and QRS widening; ASD: RBBB; Aortic stenosis: LV hypertrophy; Mitral stenosis: widened P wave, AF

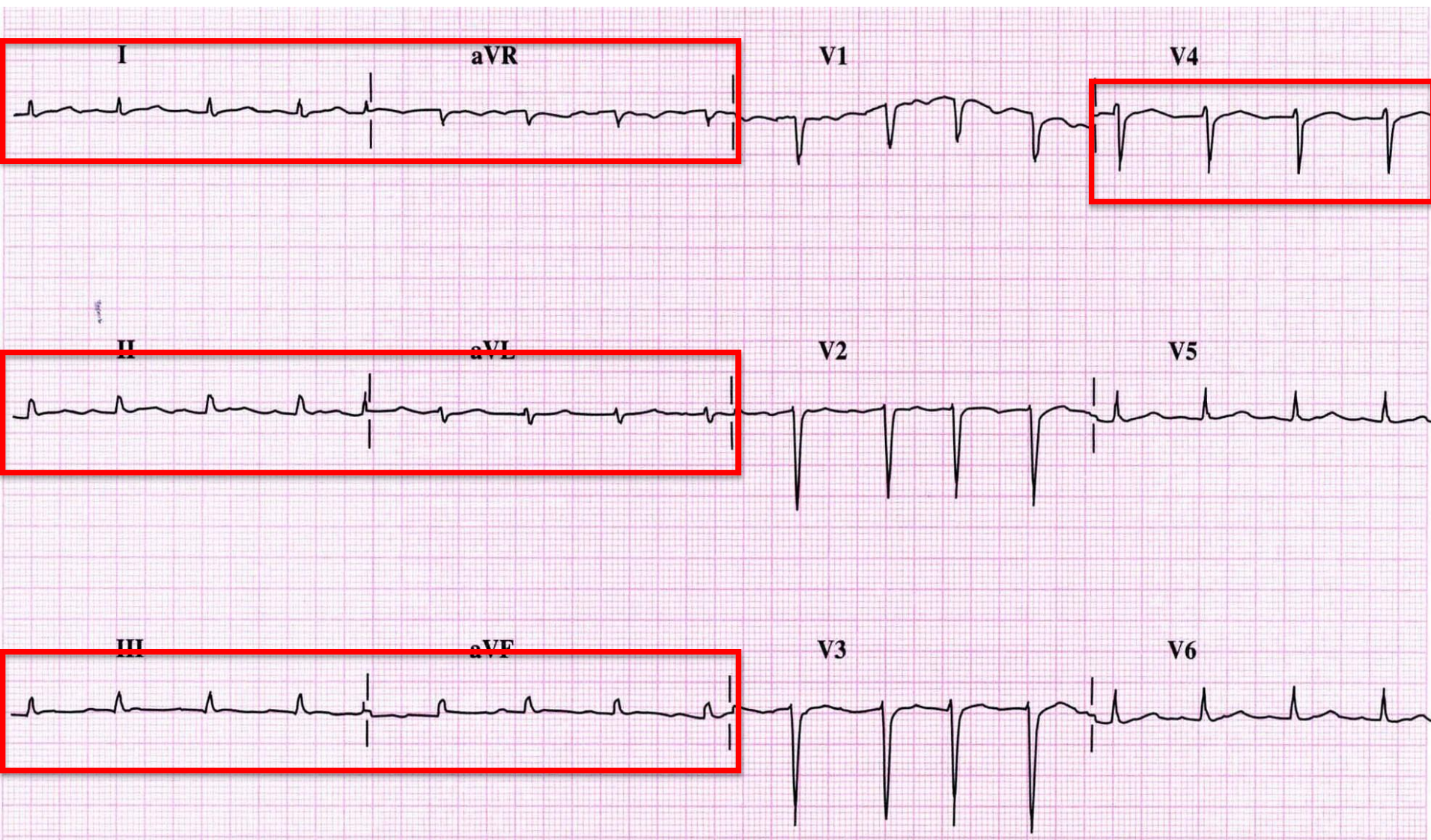
Normal ECG



- All intervals (PR, RR, QRS, QTc) and voltage are normal
- P wave: aVR -, I & II +, V1 +/-
- Axis -30° to $+90^{\circ}$ (lead I & II +/-isoelectric)
- Septal depolarization (initial 20-30ms: q waves at V5, V6, r waves at V1, V2) followed by inferolateral ventricular depolarization (next 70-80ms, transition in V3 or V4)
- No **pathological** Q waves except III and aVR
- ST segment isoelectric (except early repolarization)
- T wave upright except in AVR and V1, variable in III, aVL, aVF, V1-V2; <5mm in limb leads and <10mm in precordial leads

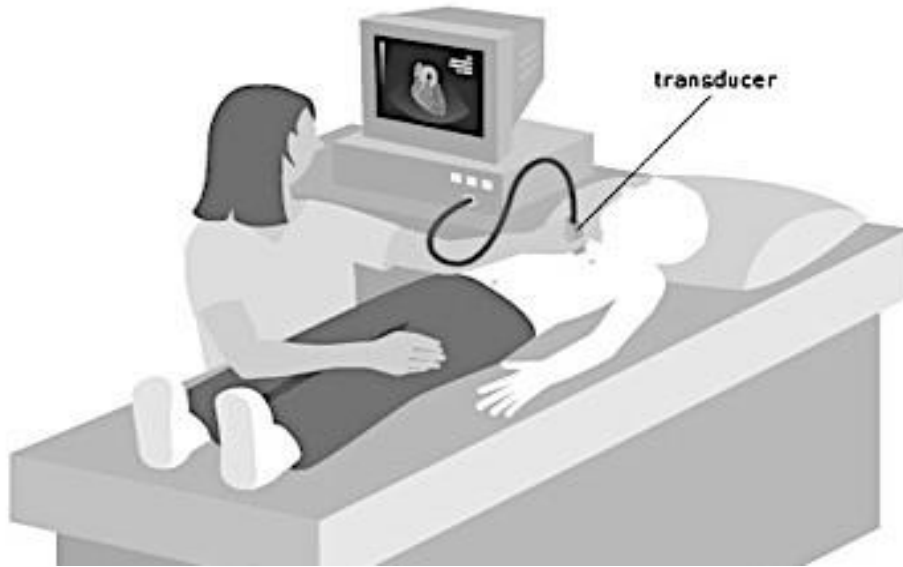






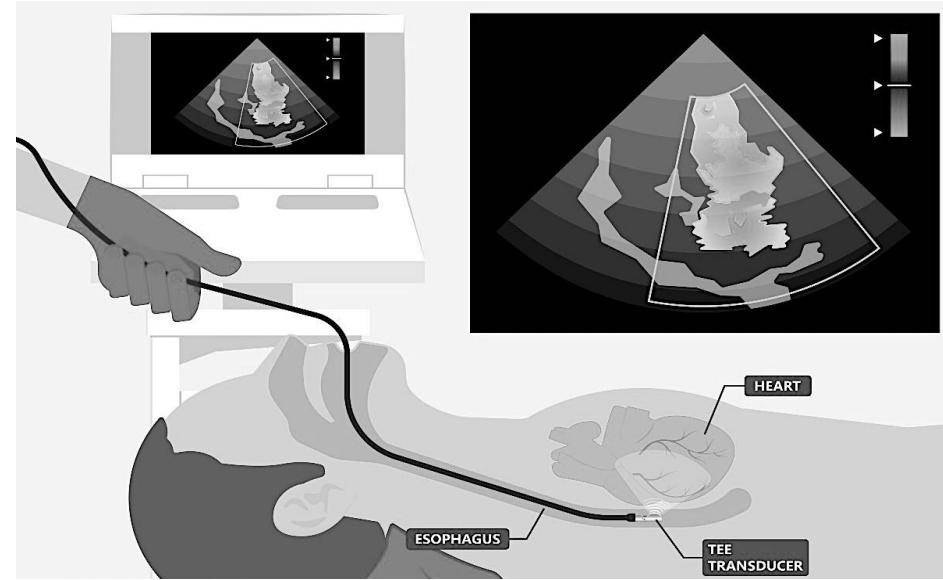
Echocardiogram

Transthoracic



1. LV thickness, chamber size, ejection fraction, diastolic function, regional wall motion abnormalities
2. RV size and systolic function
3. Valvular structure, stenosis and regurgitation
4. Right ventricular systolic pressure
5. L / R atrial size
6. Shunts (e.g. ASD, VSD)
7. Pericardial effusion
8. Aortic root size / dissection

Transesophageal



1. Valvular structure esp for vegetations and function of mechanical heart valves
2. Left atrial appendage (size, morphology, clots)
3. Shunts
4. Patent foramen ovale
5. Both ascending / descending thoracic aorta can be clearly seen

When do we offer TEE?

Risk of sedation / monitored anesthesia care

Risk of aspiration

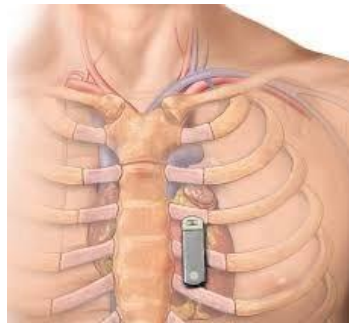
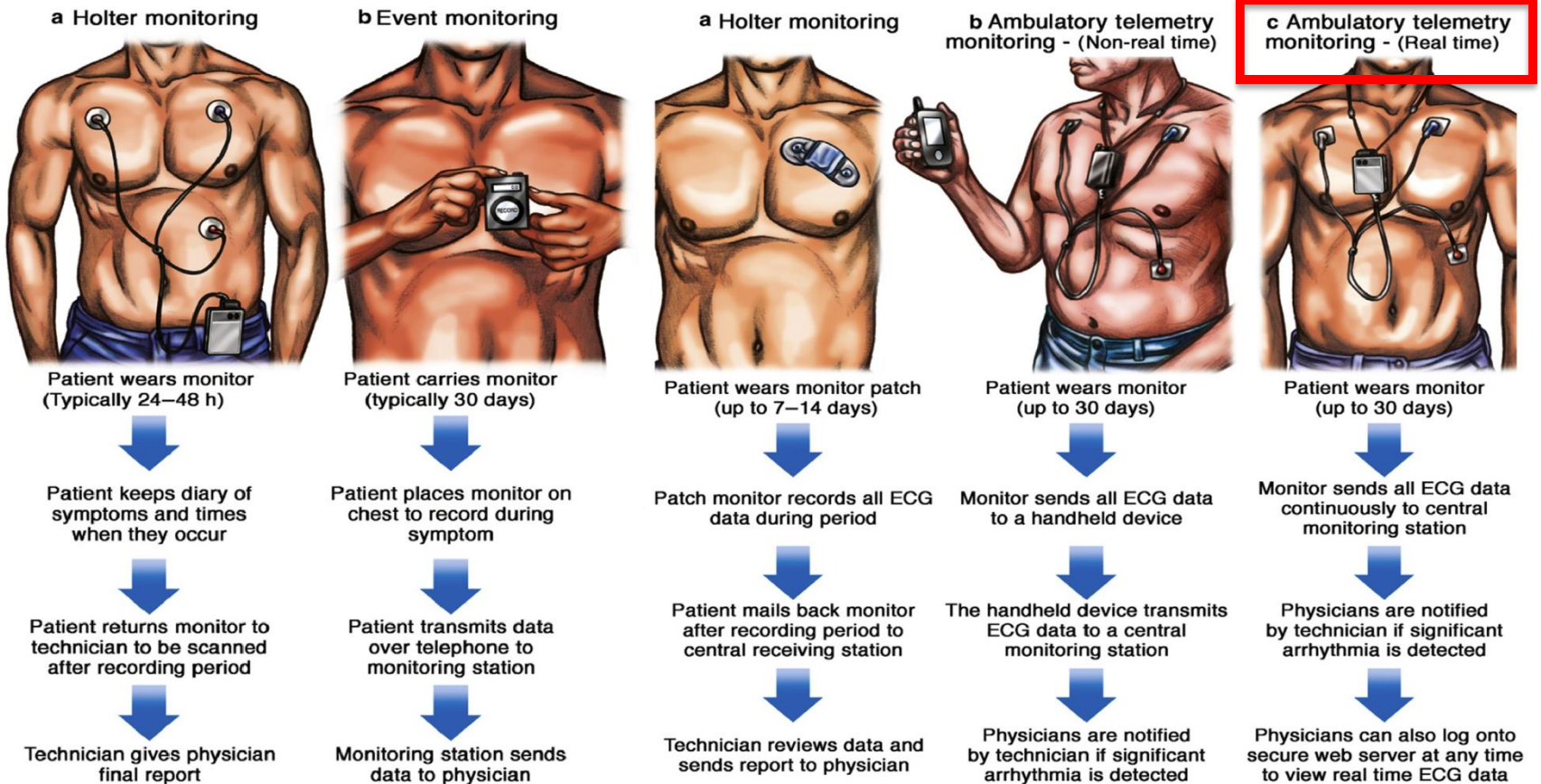
Risk of bleeding

Risk of organ perforation

Indications:

1. To diagnose (and monitor) infective endocarditis in patients with moderate or high pretest likelihood
2. To evaluate for cardiac source of embolus
3. To evaluate for acute aortic pathology (ie, dissection in unstable patients / patients who cannot tolerate contrast)
4. To evaluate for cardiac pathology when transthoracic echocardiography is nondiagnostic
5. To facilitate clinical decision-making during cardioversion / ablation for AF/Flutter
6. Assist procedures including ablation, placement of closure devices, valvuloplasty and percutaneous valvular implantation

Ambulator Cardiac Monitoring

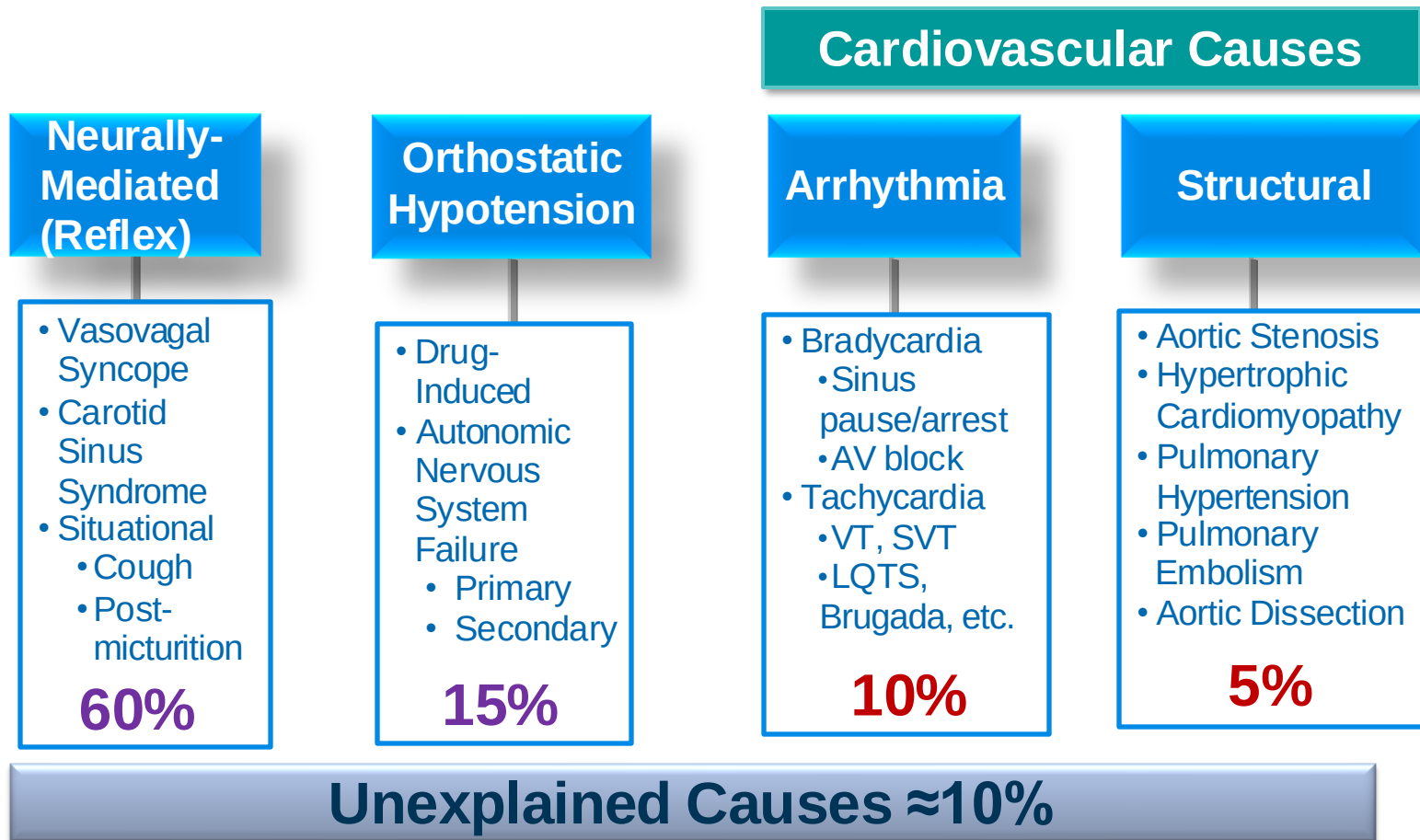


Duration of recording	<1 min	24–48 hr	3–7 days	1–4 weeks	Up to 5 years
Types of recorder	External event recorder Smartphone-based recorder	Standard Holter recorder Mobile cardiac telemetry	Patch/Vest/Belt recorder Mobile cardiac telemetry Event loop recorder	Patch/Vest/Belt recorder External loop recorder Mobile cardiac telemetry	Implantable loop recorder
Modality of recording					
Event recording	✓	✓	✓	✓	✓
Continuous recording		✓	✓	✓	
Autotriager recording			✓	✓	✓
Number of recording leads					
1 lead (2 electrodes)	✓	✓	✓	✓	✓
2 leads (3 electrodes)		✓	✓	✓	
3 leads (5–7 electrodes)		✓	✓	✓	
12 leads (10 electrodes)		✓			
Type of recording system					
Adhesive wired electrodes		✓	✓	✓	
Patch/Vest/Belt wireless system			✓	✓	
Built-in electrodes	✓				✓
Available analyses					
Arrhythmia analysis	✓	✓	✓	✓	✓
ST analysis		✓	✓	✓	
HRV—Heart rate variability		ST depression at least 0.5-1mm & at least 1 min • 60% Sn/Sp to detect CAD • In those with CAD, ½ events were likely ischemia • Better predictor of outcomes than treadmill in patients with stable CAD • Predicts short-term and long-term risk in patients with unstable CAD			
QT dynamicity					
HRT—Heart rate turbulence					
HDR—Holter-derived respiration					
QRS late potentials					
P-wave averaging					
T-wave variability					
Activity level					

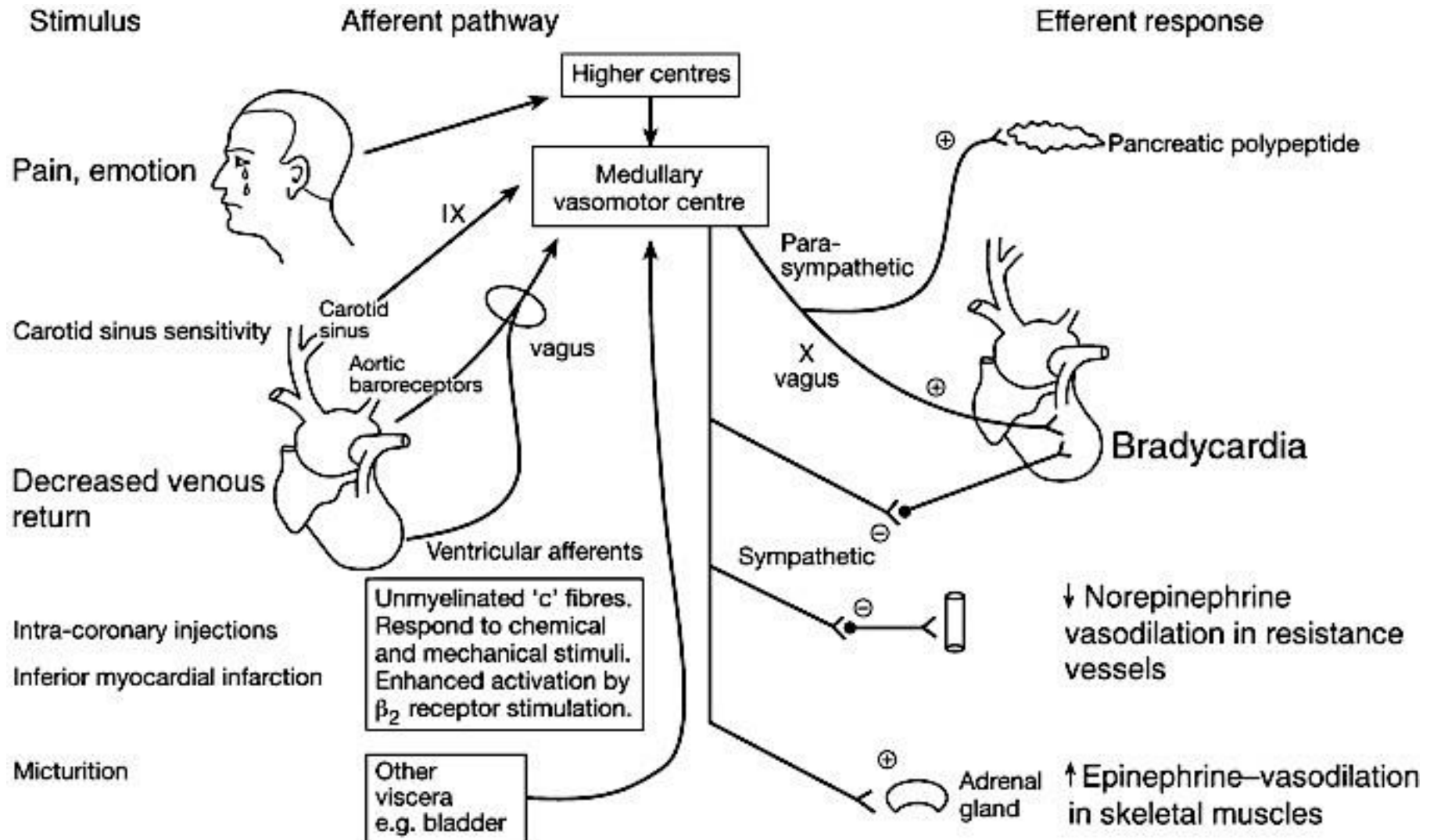
Indications for Ambulatory Cardiac Monitoring

- **Diagnosis of arrhythmias**
 - Palpitations
 - Syncope /presyncope
 - Chest pain
 - Stroke
- **Diagnosis of cardiomyopathies / primary arrhythmia syndromes**
 - *Tachycardia-mediated cardiomyopathy*: tachyarrhythmia burden
 - *Arrhythmogenic right ventricular cardiomyopathy*: VT / NSVT / PVC >500/d
 - *Long QT syndrome*: polymorphic VT
 - Uncertain diagnostic value of dynamic QRS change in Brugada / QTc change in long QT syndromes
- **Prognostication**
 - *Post MI, LVEF 35-40%*: VT/NSVT
 - *Hypertrophic cardiomyopathy*: VT/NSVT
 - *Arrhythmogenic right ventricular cardiomyopathy*: VT / NSVT / PVC >1000/d
 - *Wolff Parkinson White Syndrome*: AF, SVT, pre-excited RR interval <250ms; intermittent loss of delta wave
 - *Primary arrhythmia syndromes*: VT/VF
- **Assess treatment efficacy and side effects**

Causes of Syncope



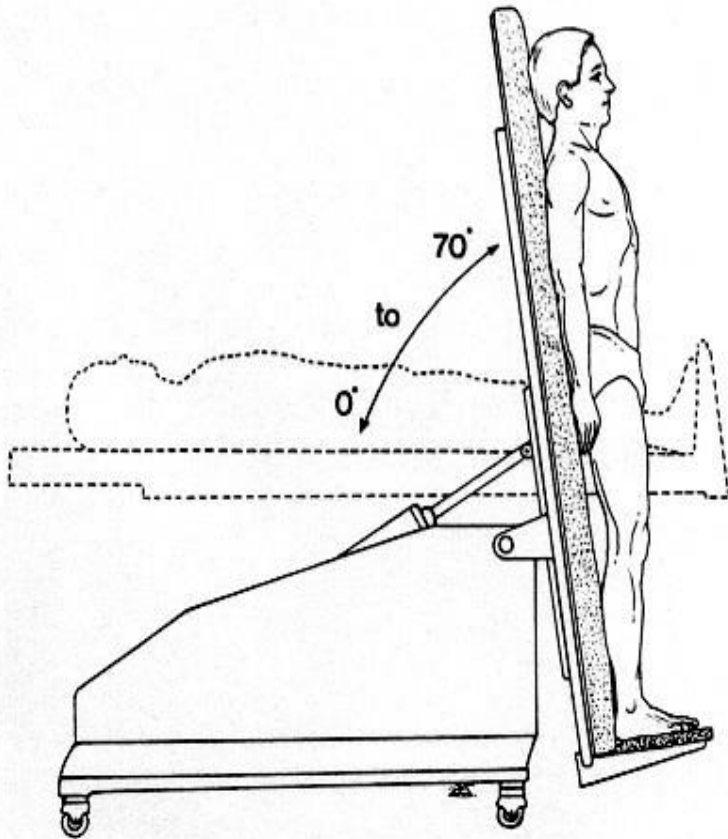
Neurally Mediated Syncope



Clinical Features of Syncope of Different Etiologies

Reflex Syncope	Orthostatic Hypotension	Cardiac Syncope
✓ Long history of recurrent syncope, especially before 40yrs old	✓ While / after standing ✓ Prolonged standing ✓ Standing after exertion ✓ Post-prandial hypotension	✓ During exertion or when supine
✓ After prolonged standing ✓ After unpleasant irritation ✓ During meal ✓ Being in crowded / hot places	✓ Started / changed the dosage of vasodepressive drugs/ diuretics ✓ Autonomic neuropathy / parkinsonism	✓ Sudden onset palpitation immediately followed by syncope
✓ Autonomic activation ✓ Head rotation / pressure on carotid sinus ✓ Absence of heart disease		✓ Family history of unexplained death at young age ✓ Presence of structural heart disease / coronary artery disease ✓ Abnormal ECG

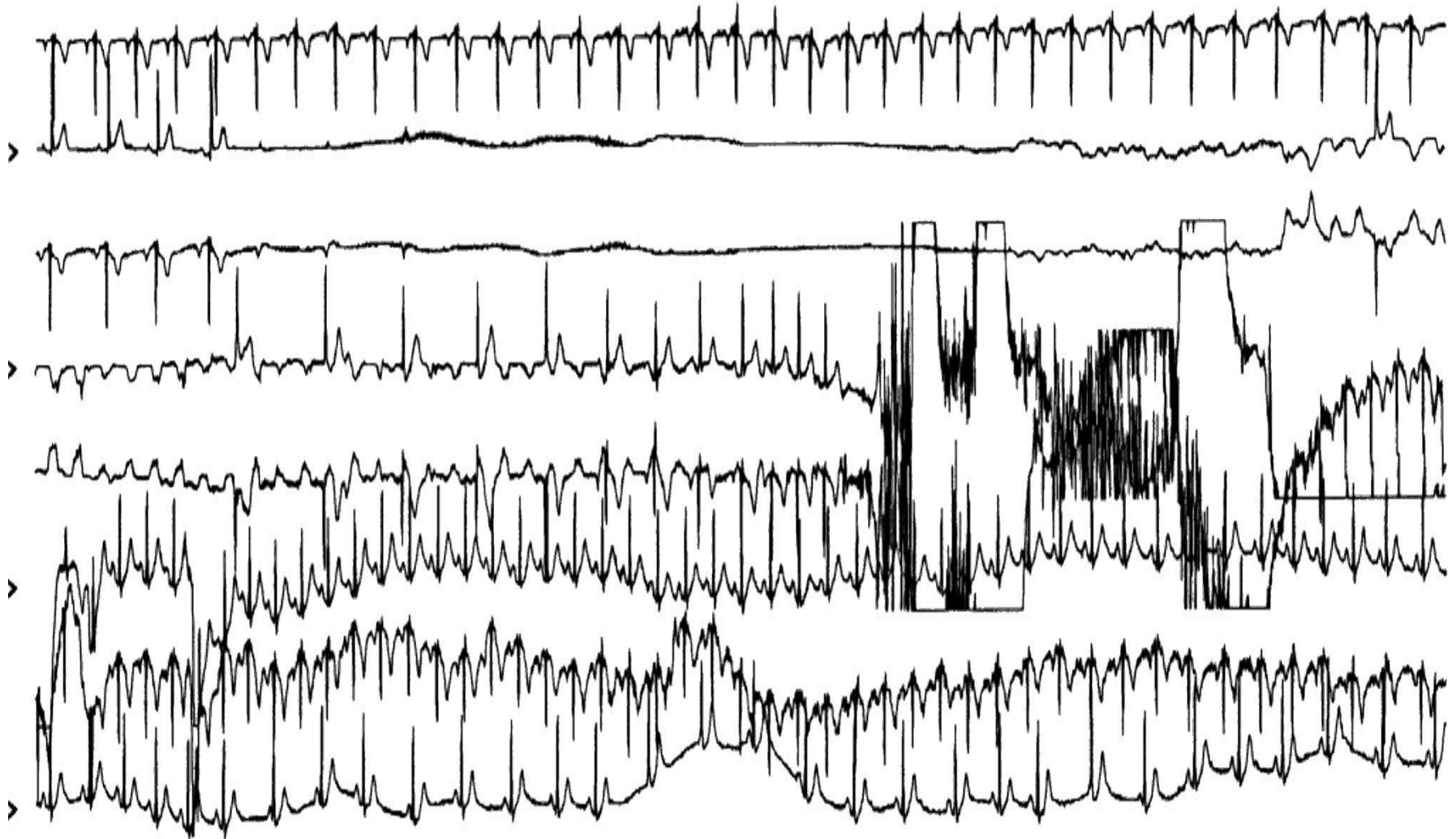
Tilt Table Test



- Tilt-table angle of 70 degrees for 30 to 40 minutes
- *Differentiate neurally-mediated syncope from orthostatic hypotension, postural orthostatic tachycardia syndrome and pseudosyncope*
- Adjunctive agents, such as a low dose of isoproterenol infusion or sublingual nitrates, may improve sensitivity but decrease specificity
- A positive tilt-table test suggests a tendency or predisposition to vasovagal syncope
- This observation during tilt-table testing cannot necessarily define a causal etiology or be entirely conclusive of a reflex mechanism for syncope in the clinical setting. Require clinical correlation with symptoms

Response/Diagnosis	Characteristics
Vasodepressor syncope	BP falls to a systolic value <60 mmHg. HR during syncope does not fall by more than 10% of its peak value.
Cardioinhibitory syncope without asystole	HR decreases <40 beats/min. for more than 10 sec; without asystole >3 sec. BP decreases before HR fall.
Cardioinhibitory syncope with asystole	Asystole >3 sec.; decrease in HR precedes or coincides with BP fall.
Mixed syncope	HR decreases during syncope but does not reach <40 beats/min., or reaches <40 beats/min. for <10 sec., with or without asystole <3 sec. BP decreases before HR fall.
Initial orthostatic hypotension	Decrease in BP >40 mmHg at standing with spontaneous and fast normalization, so that hypotension and symptoms last <30 sec.
Classic orthostatic hypotension	Decrease in systolic BP ≥ 20 mmHg and diastolic BP ≥ 10 mmHg during the first 3 min. after standing.
Late (progressive) orthostatic hypotension	Slow and progressive systolic BP decline after the 3rd min. of standing.
POTS	Increase in HR >30 beats/minutes or HR >120 beats/minute after standing, accompanied by symptoms and BP variability.
Negative TTT	Syncope, orthostatic hypotension or POTS are not provoked.

Cardioinhibition with Asystole



Advanced Imaging Modalities

- **Cardiac MRI**

- Reliable assessment of LV / **RV** volumes and systolic function
- Capability of characterizing edema (T2-STIR), scar (late gadolinium enhancement), infiltrates and subtle wall motion abnormalities
- Stress scan assess myocardial ischemia due to coronary artery disease and microvascular dysfunction

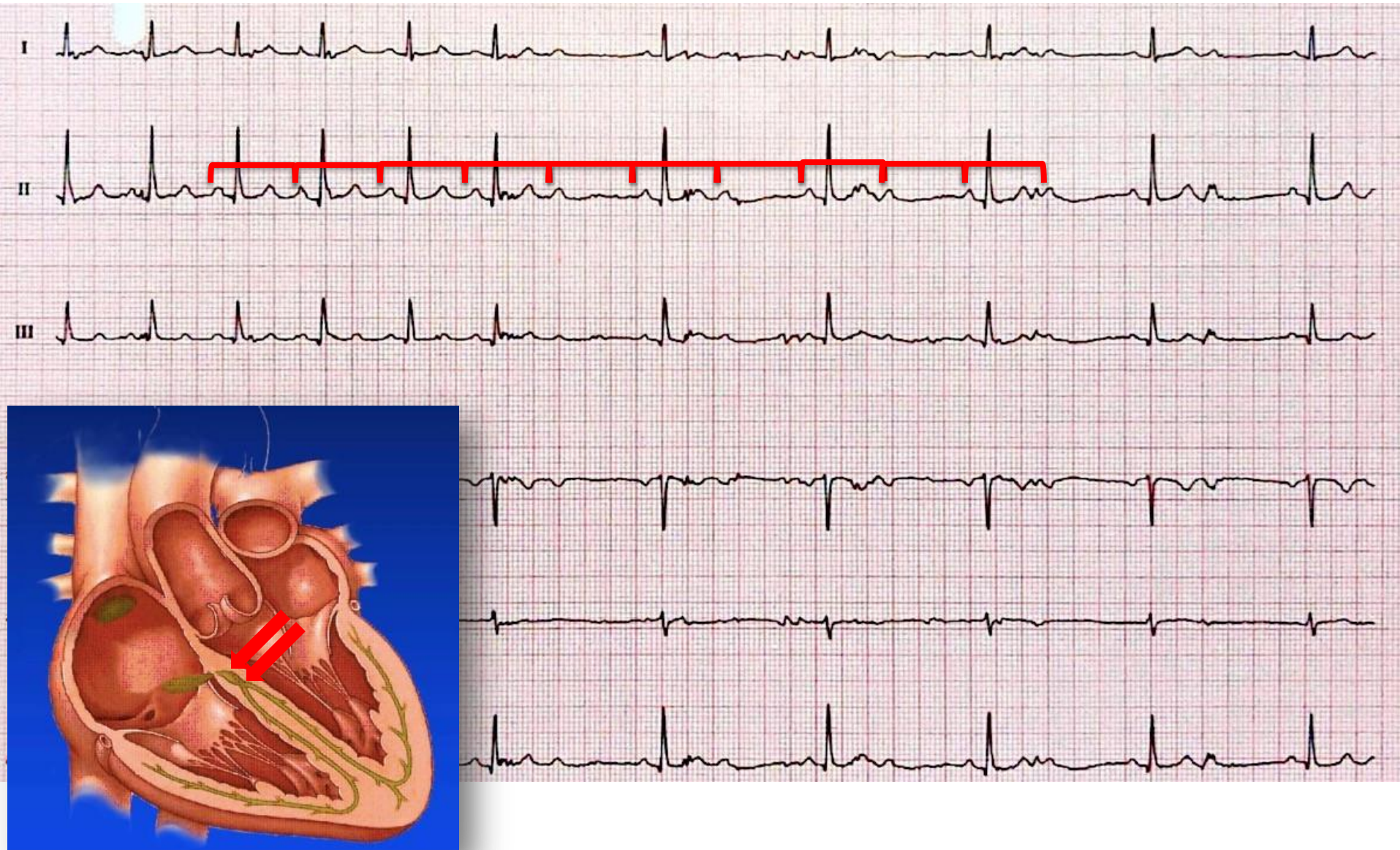
- **CT Coronary Angiography**

- **Stress Nuclear (PET / SPECT) Myocardial Perfusion Imaging**

Exercise ECG

- **Limited sensitivity / specificity for diagnosis of coronary artery disease**
- **Useful to validate exertional symptoms reported by patient**
- **Useful to assess exercise-induced arrhythmias**

Exercise ECG of a Patient with SOBOE



Others

- **Cardiac Catheterization**
 - Coronary angiography and intra-coronary imaging / hemodynamic assessment
 - Direct measurement of RA, RV, PA, LA and LV pressures
 - Indirect estimation of cardiac output and pulmonary vascular resistance
 - Remain Gold Standard for diagnosis of pulmonary artery hypertension
 - Assess presence (stepping up of oxygen saturation) and size of shunt (Q_p/Q_s)
- **Endomyocardial Biopsy**
 - Fulminant myocarditis, chronic inflammatory myocarditis, infiltrative cardiomyopathies
 - *Do it only when it affect treatment!*
- **Genetic Testing**
 - Potentially heritable cardiovascular / systemic diseases

Why do you want to perform *investigation*?

Mr Poon

- M/38
- Good past health
- 1 month history of progressive exercise intolerance
- What do you want to know?

- **Reason for exercise intolerance**

Heart Failure; Aortic Regurgitation

- Confirm the diagnosis
 - E.g. NTproBNP, CXR, echocardiogram
- Exclude differential diagnoses
 - E.g. Hb, WCC, CRP, CXR

- **Underlying cause (predisposing factors and precipitating factors)**

Marfan Syndrome; Aortic Aneurysm / Dissection

- E.g. ECG, hsTnI, CXR, echocardiogram, contrast CT thorax, genetic testing

- **Severity / complications**

Respiratory Failure; Cardiac Function; Prognosis

- E.g. NTproBNP, echocardiogram, ABG, LFT, RFT

- **Management**

Extent of Aortic Aneurysm / Dissection

- E.g. RFT as patient may need contrast CT, CT aortogram

